Fiscal Year:	FY 2007	**Task Last Updated:**	FY 06/29/2007
PI Name:	Evans, Michael L Ph.D.		
Project Title:	Probing the properties of the plant gravity sensor using a novel hypogravity simulation device		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	43210	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	02/01/2004	**End Date:**	03/31/2007
No. of Post Docs:		**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: NCE date set back to 3/31/2007 per A. Chu (4/26/07) NOTE: Received NCE to 6/30/2007 per A. Chu; previous NCE date was incorrect (4/07) NOTE: Received NCE through 9/30/2007, per A. Chu (9/2006)		
Key Personnel Changes/Previous PI:	Michael Evans		
COI Name (Institution):	Ishikawa, Hideo (Dept Plant Cellular and Molecular Biology)		
Grant/Contract No.:	NNA04CC65G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Due largely to the development and examination of gravitropism mutants and the development of new and precise means of quantitative measurement of growth and curvature, there has been substantial progress in understanding how gravity affects the growth of roots (the gravitropic growth response). However, progress in understanding gravity sensing by plants at the sub-cellular level has lagged behind. This is largely because of the lack of techniques for measuring the physical properties of the gravity receptor and relating stimulation of the receptor to the growth response. In recent years we have developed new methods for automated video analysis of plant growth ("Multi-ADAPT" and SECANT software packages) and for controlled physical manipulation of plant seedlings in a manner that allows selected gravistimulation of targeted areas ("ROTATO" plant manipulation system). While our work has continued, others have been developing and improving the Random Positioning Machine (RPM), a device designed to rotate plants randomly so as to cancel net gravistimulation and simulate microgravity conditions on earth. In this application we are merging the capabilities of the ROTATO system with the capabilities of the RPM to create a new device (the ROTATO-RPM system, or R-RPM) that will allow simulation of hypo-gravity in earth based laboratories and allow previously impossible physical manipulations of the gravity receptor in plants. The proposed research is designed to characterize the physical properties of the receptor, allow novel investigations of gravitropoism dose-response relationships, map gravisensing at the sub-cellular level, and determine the location of a recently discovered second gravity sensor. A further major outcome expected from the proposed research is the development of the R-RPM as an important tool for NASA researchers interested in the most precise simulation of hypogravity or microgravity in earth bound laboratories. The two main objectives are 1) to develop a novel hypogravity simulation device that will improve preflight testing capabilities and enhance our ability to probe the nature of the gravity detecting system in plants and 2) to use this new device to distinguish between current competing models for plant gravity sensing. With the developing budgetary constraints on life sciences research in space, there is a greatly increased need to be able to simulate flight conditions in ground based laboratories and to optimize preflight testing of experiments that will eventually be completed in space. The device that we are developing (the R-RPM) will accomplish these goals and simultaneously provide a unique tool for investigating the nature of the subcellular gravity sensor in plants, an area of research that has seen limited progress due largely to technological limitations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The more we can learn about how plants sense and respond to their environment, the more we will be able to improve our ability to enhance plant growth, productivity and survival under adverse conditions. The work proposed here will be particularly beneficial in that it will 1) lead to an overall improvement in our ability to conduct spaceflight related plant experiments (and hence futher the use of plants in long-term space flight) and 2) allow rapid advances in understanding plant growth under earth's gravitational field. The new technology being developed will be particularly beneficial in allowing more precise ground based simulation of flight leading to more economical research productivity and more rapid advances in our undrstanding of plant function.
Task Progress:	The work during the third year of funding centered on completion of the enhanced software for controlling the R and RPM components of the R-RPM, final assembly of the R-RPM, and testing the performance of the R-RPM for controlling and analyzing plant orientation under 1g and hypo-g conditions. We also aimed to test the completed R-RPM system during an analysis of gravitropism kinetics to detect possible defects remaining in the software or in the mechanical control of the system. In addition we planned to initiate studies of the mapping of gravity sensing sites in maize columella cells using the R-RPM. We accomplished our major goal of final assembly of the modified RPM and integration of the RPM with the ROTATO system to form the R-RPM. During this time we also made further improvements to our new image analysis software and then tested the software on the R-RPM to examine the performance of the image analysis software under the dynamic conditions of RPM operation. The new software proved to be excellent for our purposes. Following these accomplishments we tested the operation of the entire system in two types of analyses of early gravitropism kinetics. These projects were intended primarily to test for "bugs" in the operation of the new device. However, the data collected during these tests proved to provide unexpected insights into gravitropism motor mechanisms. The two projects were: A. Analysis of the dependence of gravitropic latent period on constant angle of stimulation and B. Analysis of the nature of the commonly observed transient reversal of curvature during maize root gravitropism. During project A , the major findings were that 1. there is an inverse relationship between stimulation angle and latent period of the gravitropic response, i.e., the smaller the stimulation angle, the quicker the response begins. This seems counter-intuitive and was unexpected. We speculate that the delayed response at high angles results from "over-stimulation" (it would be rare for a plant to suddenly shift orientation by 90o under natural conditions). To test this idea, we pre-stimulated seedlings by rotating them on the R-RPM prior to constant stimulation. Under these conditions, the response became quicker for the high angles of stimulation (as compared with non-pre-stimulated controls), suggesting that our interpretation may be correct. The significance of this finding includes the realization that typical laboratory stimulations are "abnormal" and that published latent periods of the response may have been measured under "non-natural" conditions leading to values for latent periods that are abnormally long. This has relevance for a wide variety of gravitropism analyses. During project B (transient reversal of curvature during gravitropism) the major findings were that: 1. our earlier finding of tri-phasic gravitropism (Phase 1, Straightening, Phase 2) in maize is consistent and appears under R-RPM conditions. 2. We were able to trace each phase of the response to a localized region of the root. 3. The extent of straightening is dependent upon the angle of stimulation and 4. The weaker curvature of Phase 2 continues indefinitely under conditions of constant stimulus strength. This later finding is particularly significant because it indicates that there is continuous signal output from the apex to the curving region long after any dynamic change in orientation (no signal adaptation). Examining the effects of subtle changes in angle of stimulation during Phase 2 curvature will provide useful data on signal transduction and on the true latent period between signal input and motor output. Also, it will be instructive to determine the extent to which auxin regulates Phase 1 and or Phase 2. If it only regulates Phase 2 as we suspect, the research community's thinking on the location and timing of auxin action in gravitropism needs to be re-examined. Sub-cellular mapping of gravity sensing sites in maize columella cells. This final aspect of the project was intended to be completed late during the third year of funding. However, because construction and assembly of the R-RPM took

	longer than expected, because additional time was needed to perfect the software, and because projects A and B described above were added to test the functioning of the R-RPM, the planned mapping experiments were not completed. However, we now have the very specialized technology required to carry out these experiments in the future.
Bibliography Type:	Description: (Last Updated: 03/08/2019)
Abstracts for Journals and Proceedings	Ishikawa I, Natori E, Evans ML. "Use of rotato/random positioning machine (R/RPM) technology to investigate gravity sensing and the gravitropic motor response of maize roots. " Presented at the Annual Meeting of the American Society for Gravitational and Space Biology, Arlington, Virginia, November 2-6, 2006. Gravitational and Space Biology 2006 Nov; 20(1):27. , Nov-2006
Abstracts for Journals and Proceedings	Natori E, Evans ML, Ishikawa H. "Analysis of the triphasic motor response in maize root gravitropism. " Presented at the Annual Meeting of the American Society for Gravitational and Space Biology, Arlington, Virginia, November 2-5, 2006. Gravitational and Space Biology 2006 Nov; 20(1):27. , Nov-2006
Articles in Peer-reviewed Journals	Ishikawa H, Natori E, Evans M. "Use of rotato/random positioning machine (R/RPM) technology to investigate gravity sensing and the gravitropic motor response of maize roots." Gravitational and Space Biology, 2007 Jun;20(2):101-2. , Jun-2007